

Python Assessment - Beginner to Intermediate

Question 1: Student Result Analyzer (10 Marks)

Create a function that accepts a list of student marks.

1. Calculate the total marks.
2. Calculate the average marks.
3. Find the highest and lowest marks.
4. Count how many students passed (marks ≥ 40).
5. Return all the calculated values.

Example: marks = [78, 45, 32, 91, 67, 39]

Expected Output:

Total Marks: 352
Average Marks: 58.67
Highest Marks: 91
Lowest Marks: 32
Passed Students: 4

Question 2: Shopping Cart Calculator (10 Marks)

Create a function that accepts a list of item prices and a discount percentage (default = 10).

1. Calculate the total bill.
2. Apply the discount.
3. If the final bill is greater than ₹5000, apply an additional 5% discount.
4. Return the final payable amount.

Example: prices = [1200, 800, 1500, 2500]

Expected Output:

Original Bill: 6000
Discount Applied: 10%
Extra Discount: Yes
Final Amount: 5130

Question 3: Employee Information System (10 Marks)

Create a function using ****kwargs** to store employee details.

1. Print all employee details.
2. If salary is greater than ₹50,000, print 'High Salary', otherwise print 'Normal Salary'.
3. Count the total number of details received.
4. Return the total count.

Example:

employee(name='Rahul', age=25, department='IT', salary=60000)

Expected Output:

Name: Rahul
Age: 25
Department: IT
Salary: 60000
High Salary
Total Details: 4

Question 4: Number Analysis Tool (20 Marks)

Create a function that accepts any number of integers using ***args**.

1. Find the largest and smallest number.
2. Count even and odd numbers.
3. Calculate the sum and average.
4. Print all numbers divisible by both 3 and 5.
5. Return all calculated values.

Example: `analyze_numbers(12, 15, 20, 9, 30, 7, 4)`

Expected Output:

Largest Number: 30

Smallest Number: 4

Even Numbers: 4

Odd Numbers: 3

Sum: 97

Average: 13.86

Divisible by both 3 and 5: 15, 30